

4.5 Negative Exponents and Reciprocals



LESSON FOCUS

Relate negative exponents to reciprocals.

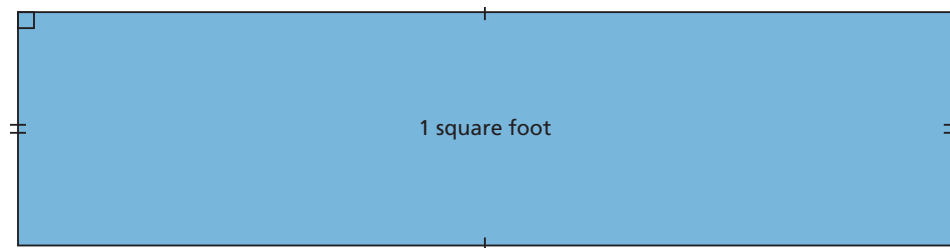
Scientists can calculate the speed of a dinosaur from its tracks. These tracks were found near Grand Cache, Alberta.

Make Connections

A rectangle has area 1 square foot.

List 5 possible pairs of lengths and widths for this rectangle.

What is the relationship between the possible lengths and widths?



Construct Understanding

TRY THIS

Work with a partner.

You will need grid paper and scissors.

- A.** Cut out a 16 by 16 grid. Determine the area of the grid in square units and as a power of 2. Record your results in a table like this:

Cut	Area (units ²)	Area as a Power of 2
Start	256	
1		
2		
3		

- B.** Cut the grid in half and discard one piece.
In the table, record the area of the remaining piece in square units and as a power of 2.
- C.** Repeat Step B until the paper cannot be cut further.
- D.** Use patterns to extend the second and third columns of the table to Cut 13.
- E.** Compare the areas for each pair of powers in the table:
■ 2^{-1} and 2^1 ■ 2^{-2} and 2^2 ■ 2^{-3} and 2^3
What relationships do you notice?

Two numbers with a product of 1 are reciprocals.

Since $4 \cdot \frac{1}{4} = 1$, the numbers 4 and $\frac{1}{4}$ are reciprocals.

Similarly, $\frac{2}{3} \cdot \frac{3}{2} = 1$, so the numbers $\frac{2}{3}$ and $\frac{3}{2}$ are also reciprocals.

We define powers with negative exponents so that previously developed properties such as $a^m \cdot a^n = a^{m+n}$ and $a^0 = 1$ still apply.

Apply these properties.

$$\begin{aligned} 5^{-2} \cdot 5^2 &= 5^{-2+2} \\ &= 5^0 \\ &= 1 \end{aligned}$$

Since the product of 5^{-2} and 5^2 is 1, 5^{-2} and 5^2 are reciprocals.

$$\text{So, } 5^{-2} = \frac{1}{5^2} \quad \text{and} \quad \frac{1}{5^{-2}} = 5^2$$

$$\text{That is, } 5^{-2} = \frac{1}{25}$$

This suggests the following definition for powers with negative exponents.

Powers with Negative Exponents

When x is any non-zero number and n is a rational number, x^{-n} is the reciprocal of x^n .

That is, $x^{-n} = \frac{1}{x^n}$ and $\frac{1}{x^{-n}} = x^n$, $x \neq 0$

Why can't x be 0?

Example 1 Evaluating Powers with Negative Integer Exponents

Evaluate each power.

a) 3^{-2} b) $\left(-\frac{3}{4}\right)^{-3}$ c) 0.3^{-4}

SOLUTION

a) $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$ b) $\left(-\frac{3}{4}\right)^{-3} = \left(-\frac{4}{3}\right)^3 = -\frac{64}{27}$ c) 0.3^{-4}

Use a calculator.

$0.3^{-4} = 123.4567901$

$0.3^{-4} = 123.4567\dots$

CHECK YOUR UNDERSTANDING

1. Evaluate each power.

a) 7^{-2} b) $\left(\frac{10}{3}\right)^{-3}$

c) $(-1.5)^{-3}$

[Answers: a) $\frac{1}{49}$ b) $\frac{27}{1000}$

c) $-0.2962\dots$]

We can apply the meaning of rational exponents and negative exponents to evaluate powers with negative rational exponents. For example, the rational exponent in the power $8^{-\frac{2}{3}}$ indicates these operations at the right.

reciprocal
↓
 $\frac{2}{3}$ ← square
8 ← cube root

Since the exponent $-\frac{2}{3}$ is the product: $(-1)\left(\frac{1}{3}\right)(2)$, and order does not matter when we multiply, we can apply the three operations of reciprocal, cube root, and square in any order.

Example 2 Evaluating Powers with Negative Rational Exponents

Evaluate each power without using a calculator.

a) $8^{-\frac{2}{3}}$ b) $\left(\frac{9}{16}\right)^{-\frac{3}{2}}$

(Solution continues.)

SOLUTION

a) $8^{-\frac{2}{3}} = \frac{1}{8^{\frac{2}{3}}}$ Write with a positive exponent.

$= \frac{1}{(\sqrt[3]{8})^2}$ Take the cube root.

$= \frac{1}{2^2}$ Square the result.

$= \frac{1}{4}$

b) $\left(\frac{9}{16}\right)^{-\frac{3}{2}} = \left(\frac{16}{9}\right)^{\frac{3}{2}}$ Write with a positive exponent.

$= \left(\sqrt{\frac{16}{9}}\right)^3$ Take the square root.

$= \left(\frac{4}{3}\right)^3$ Cube the result.

$= \frac{64}{27}$

CHECK YOUR UNDERSTANDING

2. Evaluate each power without using a calculator.

a) $16^{-\frac{5}{4}}$ b) $\left(\frac{25}{36}\right)^{-\frac{1}{2}}$

[Answers: a) $\frac{1}{32}$ b) $\frac{6}{5}$]

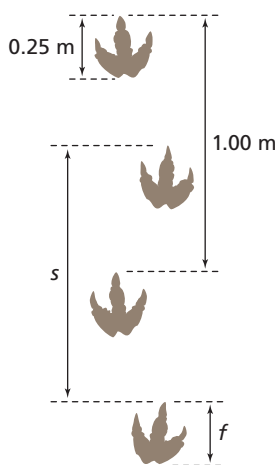
What other strategies could you use to evaluate the powers?

Example 3 Applying Negative Exponents

Paleontologists use measurements from fossilized dinosaur tracks and the formula

$v = 0.155 s^{\frac{5}{3}} f^{-\frac{7}{6}}$ to estimate the speed at which the dinosaur travelled. In the formula, v is the speed in metres per second, s is the distance between successive footprints of the same foot, and f is the foot length in metres.

Use the measurements in the diagram to estimate the speed of the dinosaur.



SOLUTION

Use the formula: $v = 0.155 s^{\frac{5}{3}} f^{-\frac{7}{6}}$

Substitute: $s = 1$ and $f = 0.25$

$v = 0.155 (1)^{\frac{5}{3}} (0.25)^{-\frac{7}{6}}$

$v = 0.155 (0.25)^{-\frac{7}{6}}$

$v = 0.7811\dots$

The dinosaur travelled at approximately 0.8 m/s.

$0.155(0.25)^{-7/6}$
0.781151051

CHECK YOUR UNDERSTANDING

3. Use the formula $v = 0.155 s^{\frac{5}{3}} f^{-\frac{7}{6}}$ to estimate the speed of a dinosaur when $s = 1.5$ and $f = 0.3$.

[Answer: approximately 1.2 m/s]

What is the speed of the dinosaur in kilometres per hour?

Discuss the Ideas

- When m is an integer, describe the relationship between a^m and a^{-m} .
- Why is there usually more than one way to determine the value of a power of the form $a^{-\frac{m}{n}}$? Use examples to justify your answer.

Exercises

A

- Copy then complete each equation.

a) $\frac{1}{5^4} = 5^{\square}$	b) $\left(-\frac{1}{2}\right)^{-3} = (-2)^{\square}$
c) $\frac{1}{3^{\square}} = 3^2$	d) $\frac{1}{4^{-2}} = 4^{\square}$
- Evaluate the powers in each pair without a calculator.

a) 4^2 and 4^{-2}	b) 2^4 and 2^{-4}
c) 6^1 and 6^{-1}	d) 4^3 and 4^{-3}

 Describe what is similar about the answers, and what is different.
- Given that $2^{10} = 1024$, what is 2^{-10} ?
- Write each power with a positive exponent.

a) 2^{-3}	b) 3^{-5}	c) $(-7)^{-2}$
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- Write each power with a positive exponent.

a) $\left(\frac{1}{2}\right)^{-2}$	b) $\left(\frac{2}{3}\right)^{-3}$	c) $\left(-\frac{6}{5}\right)^{-4}$
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- Evaluate each power without using a calculator.

a) 3^{-2}	b) 2^{-4}	c) $(-2)^{-5}$
d) $\left(\frac{1}{3}\right)^{-3}$	e) $\left(-\frac{2}{3}\right)^{-2}$	f) $\frac{1}{5^{-3}}$

B

- Evaluate each power without using a calculator.

a) $4^{-\frac{1}{2}}$	b) $0.09^{-\frac{1}{2}}$
c) $27^{-\frac{1}{3}}$	d) $(-64)^{-\frac{1}{3}}$
e) $(-0.027)^{-\frac{2}{3}}$	f) $32^{-\frac{2}{5}}$
g) $9^{-\frac{3}{2}}$	h) $0.04^{-\frac{3}{2}}$
- Use a power with a negative exponent to write an equivalent form for each number.

a) $\frac{1}{9}$	b) $\frac{1}{5}$	c) 4	d) -3
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- When you save money in a bank, the bank pays you *interest*. This interest is added to your investment and the resulting amount also earns interest. We say the interest *compounds*. Suppose you want an amount of \$3000 in 5 years. The interest rate for the savings account is 2.5% compounded annually. The money, P dollars, you must invest now is given by the formula: $P = 3000(1.025)^{-5}$. How much must you invest now to have \$3000 in 5 years?
- Here is a student's solution for evaluating a power. Identify any errors in the solution. Write a correct solution.

$$\begin{aligned} \left(-\frac{64}{125}\right)^{-\frac{5}{3}} &= \left(\frac{64}{125}\right)^{\frac{5}{3}} \\ &= \left(\sqrt[3]{\frac{64}{125}}\right)^5 \\ &= \left(\frac{4}{5}\right)^5 \\ &= \frac{1024}{3125} \end{aligned}$$

- Evaluate each power without using a calculator.

a) $27^{-\frac{4}{3}}$	b) $16^{-1.5}$	c) $32^{-0.4}$
d) $\left(-\frac{8}{27}\right)^{-\frac{2}{3}}$	e) $\left(\frac{81}{16}\right)^{-\frac{3}{4}}$	f) $\left(\frac{9}{4}\right)^{-\frac{5}{2}}$
- Michelle wants to invest enough money on January 1st to pay her nephew \$150 at the end of each year for the next 10 years. The savings account pays 3.2% compounded annually. The money, P dollars, that Michelle must invest today is given by the formula $P = \frac{150[1 - 1.032^{-10}]}{0.032}$. How much must Michelle invest on January 1st?

15. The intensity of light at its source is 100%.
The intensity, I , at a distance d centimetres from the source is given by the formula $I = 100d^{-2}$.
Use the formula to determine the intensity of the light 23 cm from the source.
16. Which is greater, 2^{-5} or 5^{-2} ? Verify your answer.
17. a) Identify the patterns in this list.
 $16 = 2^4$
 $8 = 2^3$
 $4 = 2^2$
 b) Extend the patterns in part a downward.
 Write the next 5 rows in the pattern.
 c) Explain how this pattern shows that $a^{-n} = \frac{1}{a^n}$.
18. How many times as great as 3^{-5} is 3^3 ? Express your answer as a power and in standard form.
19. What do you know about the sign of the exponent in each case? Justify your answers.
 a) $3^x > 1$ b) $3^x < 1$ c) $3^x = 1$

C

20. A number is raised to a negative exponent. Is it always true that the value of the power will be less than 1? Use an example to explain.
21. There is a gravitational force, F newtons, between Earth and the moon. This force is given by the formula $F = (6.67 \times 10^{-11})Mmr^{-2}$, where M is the mass of Earth in kilograms, m is the mass of the moon in kilograms, and r is the distance between Earth and the moon in metres. The mass of Earth is approximately 5.9736×10^{24} kg. The mass of the moon is approximately 7.349×10^{22} kg. The mean distance between them is approximately 382 260 km.
- a) What is the gravitational force between Earth and the moon?
- b) The value r is actually the distance between the centres of Earth and the moon. Research to find the diameters of Earth and the moon. Calculate the gravitational force with this new value of r .

Reflect

Explain what a negative exponent means. Use examples to demonstrate your thinking.



THE WORLD OF MATH

Math Fact: Computers and π

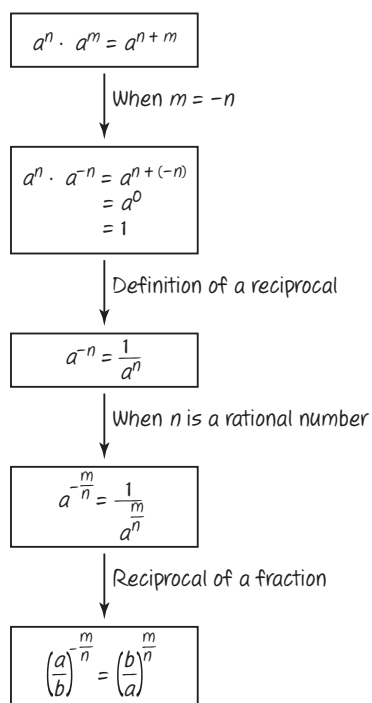
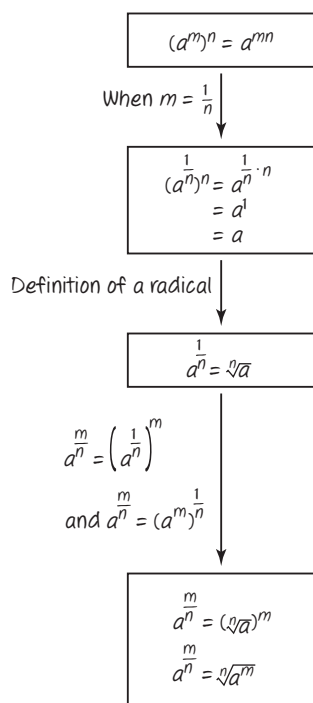
The most famous irrational number of all, π , was not proved irrational until the 18th century. Throughout history, many mathematicians have spent much time manually calculating the decimal digits of π . This had no practical value, but did provide motivation for some discoveries that have been influential in modern mathematics. In 1949, an early computer calculated π to 2000 digits, which was many more digits than any previous manual calculations. Today, calculating digits of π is a test of the power, speed, and accuracy of supercomputers. Computers have calculated π to more than 6 billion digits, and even a home computer can be programmed to calculate π to millions of digits in a few hours. This T-shirt shows the first 100 digits of π .





CHECKPOINT 2

Connections



Concept Development

In Lesson 4.4

- You used patterns to explain a meaning for an exponent of the form $\frac{1}{n}$.
- You applied the exponent law for the power of a power to justify why a **power with exponent $\frac{1}{n}$ is the n th root of the base of the power.**

In Lesson 4.5

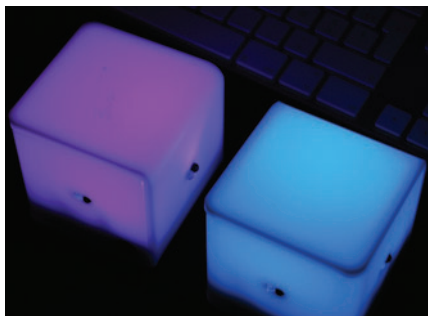
- You used patterns to explain a meaning for a negative exponent.
- You applied the exponent law for multiplying powers to justify why a **power with a negative rational exponent is written as a reciprocal.**

Assess Your Understanding

4.4

- Evaluate each power without using a calculator.
 - $16^{\frac{1}{4}}$
 - $49^{0.5}$
 - $(-64)^{\frac{2}{3}}$
 - $\left(\frac{49}{9}\right)^{1.5}$
 - $(-8)^{\frac{5}{3}}$
- Write each power as a radical.
 - $35^{\frac{2}{3}}$
 - $32^{\frac{3}{2}}$
 - $(-32)^{\frac{2}{5}}$
 - $400^{1.5}$
 - $(-125)^{\frac{1}{3}}$
 - $\left(\frac{8}{125}\right)^{\frac{2}{3}}$
 - Evaluate each radical in part a without using a calculator, if possible. Explain why you could not evaluate some radicals.
- Write each radical in exponent form.
 - $\sqrt[3]{4}$
 - $\sqrt{9}$
 - $\sqrt[4]{18}$
 - $(\sqrt{10})^3$
 - $(\sqrt[3]{-10})^2$
- The circulation time is the average time it takes for all the blood in the body to circulate once and return to the heart. The circulation time for a mammal can be estimated from the formula $T \doteq 17.4m^{\frac{1}{4}}$, where T is the circulation time in seconds and m is the body mass in kilograms. Estimate the circulation time for a mammal with mass 85 kg.
- Arrange these numbers in order from least to greatest.

$$3^{\frac{3}{2}}, \sqrt[3]{3}, (\sqrt{3})^5, 3^{\frac{2}{3}}, (\sqrt[3]{3})^4$$
- An AudioCube creates sounds and musical patterns. Each cube exchanges musical information wirelessly with nearby cubes. The volume of an AudioCube is $421\,875 \text{ mm}^3$. Write the edge length of the cube as a radical and as a power, then calculate the edge length.



4.5

- Evaluate each power without using a calculator.
 - $\left(\frac{2}{3}\right)^{-4}$
 - 0.5^{-2}
 - $(-1000)^{-\frac{2}{3}}$
 - $\left(\frac{1}{4}\right)^{-\frac{1}{2}}$
 - $\left(\frac{1}{10}\right)^{-2}$
 - $(-0.008)^{-\frac{4}{3}}$
- Suppose you want \$5000 in 3 years. The interest rate for a savings account is 2.9% compounded annually. The money, P dollars, you must invest now is given by the formula: $P = 5000(1.029)^{-3}$. How much must you invest now to have \$5000 in 3 years?